Implementação de Machine Learning com Prophet

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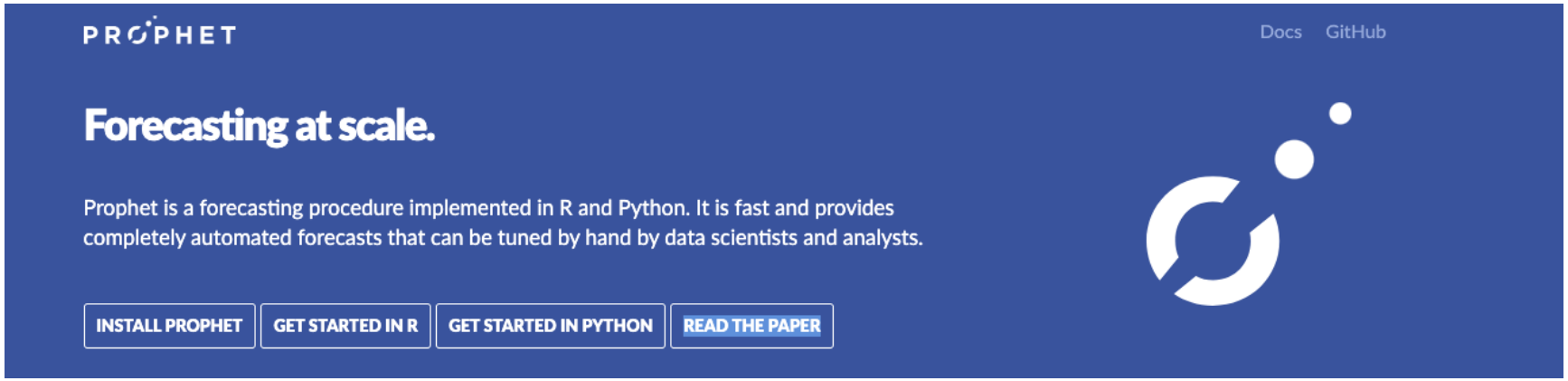
Definição



https://facebook.github.io/prophet/

Procedimentos de Uso

PROPHET

Docs GitHub

Documentation

Installation
- Using R
- Using Python

Quick Start
- Python API
- R API

Saturating Forecasts
- Forecasting Growth
- Saturating Minimum

Trend Changepoints
- Automatic changepoint detection in Prophet
- Adjusting trend flexibility
- Specifying the locations of the changepoints

Seasonality, Holiday Effects, And Regressors
- Modeling Holidays and Special Events
- Built-in Country

Quick Start

Python API

Prophet follows the `sklearn` model API. We create an instance of the `Prophet` class and then call its `fit` and `predict` methods.

The input to Prophet is always a dataframe with two columns: `ds` and `y`. The `ds` (datestamp) column should be of a format expected by Pandas, ideally YYYY-MM-DD for a date or YYYY-MM-DD HH:MM:SS for a timestamp. The `y` column must be numeric, and represents the measurement we wish to forecast.

As an example, let's look at a time series of the log daily page views for the Wikipedia page for Peyton Manning. We scraped this data using the Wikipediatrend package in R. Peyton Manning provides a nice example because it illustrates some of Prophet's features, like multiple seasonality, changing growth rates, and the ability to model special days (such as Manning's playoff and superbowl appearances). The CSV is available here.

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Forecasting at scale

Data Science

Sean J Taylor*, Benjamin Letham*

September 27, 2017

Highlighted in Practical Data Science for Stats

Author and article information

Abstract

Forecasting is a common data science task that helps organizations with capacity planning, goal setting, and anomaly detection. Despite its importance, there are serious challenges associated with producing reliable and high quality forecasts — especially when there are a variety of time series and analysts with expertise in time series modeling are relatively rare. To address these challenges, we describe a practical approach to

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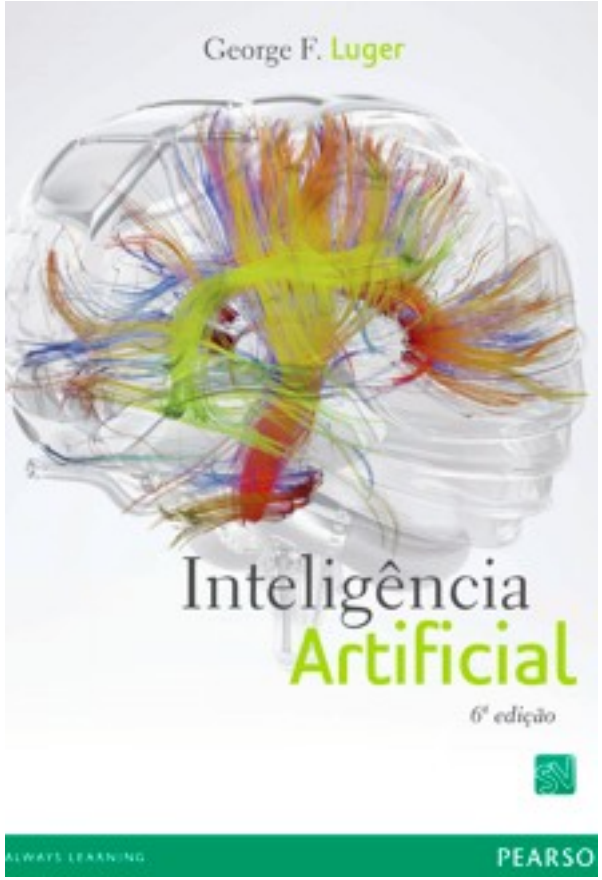

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